DARDEN CASEBOOK SUPPLEMENT 2005

This year the Consulting Club with the help of volunteers collected a selection of actual cases that Darden students experienced during their case interviews with major consulting firms, either for their internship or full-time position. We hope that this casebook supplement will be a valuable resource to you in your interview preparation.

A big thank you goes to all the contributors of the cases presented:

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Good luck with your job-search process and we hope that when you receive actual cases you will pay it forward as well by contributing to the next version of this casebook supplement.

Angela Huang
Editor and VP of Education, Consulting Club
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Actual Case #1
Company: AT Kearney

Case Description
A profitability/market estimation case that included industry analysis components, as well.

Question
The leading cookie manufacturer in the United States has contacted us because they are concerned about the growth of the private label cookie business.

Recently, the private label market has grown substantially and taken overall market share away from the brand name cookie manufacturers.

One of our client's major competitors has recently entered the private label market, and the client is deciding whether or not they should do the same.

Initial data to be given
Candidate should be asked to first estimate the size of the cookie market in the United States (in dollars). They should also figure out how many players are in the market - there is one major competitor and several smaller players. Specifics about the cookie manufacturing industry can be provided if asked for, but are not critical to the case.

Additional data to be provided when asked
Consumer preferences have shifted in recent years and people are more price sensitive in the cookie market than they used to be. This is why the private label market has grown, stealing share from the branded cookie manufacturers.

Overall the market for cookies in the U.S. has remained steady for the past 5 years. Market share data for the US Cookie industry is below and can be handed to the interviewee.

Frameworks that might be used
Internal/external analysis, 5C's (Company, Customers, Collaborators, Competitors, and Context).

Solution
The candidate should properly size the market in the neighborhood of $1B and calculate total revenue figures from the data provided on the worksheet. They should figure out what the potential opportunity is in the private label cookie market and estimate how much of that revenue the client could potentially capture. Realizing that the main competitor who entered the space would have lost much more revenue had they not entered the private label market, the interviewee should probably recommend that the client participate, as well. In terms of challenges the company might face by entering the market, it would be good to mention cannibalization of existing brand sales, the lower margins received on the private label cookies, commoditization of the marketplace, and cultural issues within the company that could arise from producing a lower quality product.

Candidate's Copy

Market Share Data for U.S. Cookie Industry			
(millions of $)			
	5 years ago	**3 years ago**	**Today**
Overall Cookie Market			
Overall Market Size ($M)	$1,000	$1,000	$1,000
Brand as % of Total Market	100%	90%	80%
Private Label as % of Total Market	0%	10%	20%
Brand Market Share			
Client Share of Brand Market	60%	67%	70%
Main Competitor Share of Brand Market	30%	25%	23%
Other Players Share of Brand Market	10%	8%	7%
Private Lable Market Share			
Client Share of Private Label Market	0%	0%	0%
Main Competitior Share of Private Label Market	0%	0%	40%
Other Players Share of Private Label Market	0%	100%	60%

Actual Case #2
Company: BCG

Case Description
A profitability case, with industry analysis components.

Question
Our client sells scooters and motorcycles in an emerging-market country. Recently, revenue and profits have been declining.

What might be causing the dip in revenue?

What can the client do to reverse this trend?

Initial data to be given
Customers are typically lower-income consumers who cannot afford a car.

Customers use the scooters to get to work and get around for everyday tasks.

Additional data to be provided when asked
(Give the full sheet of data to the candidate)

Scooter Data		
	5 yrs ago	**Today**
Company Data		
Market Size (# motorcycles)	1,000,000	900,000
Market Share	70%	70%
Average Retail Price	$500	$500
Fixed Costs (per scooter)	$200	$200
Variable Costs (per scooter)	$200	$200
Customer Data		
Salvage Value	$0	$0
Average Useful Life (years)	4	4
Gas Mileage (mpg)	60	60
Average Annual Maintenance Cost	$100	$100

Motorcycle Data		
	5 yrs ago	**Today**
Company Data		
Market Size (# motorcycles)	100,000	500,000
Market Share	10%	10%
Average Retail Price	$600	$500
Fixed Costs (per scooter)	$200	$200
Variable Costs (per scooter)	$200	$200
Customer Data		
Salvage Value	$50	$50
Average Useful Life (years)	4	4
Gas Mileage (mpg)	60	60
Average Annual Maintenance Cost	$100	$100

The company has not pursued any expansion in the motorcycle category and would prefer to improve on the scooter side.

Frameworks that might be used

Calculate the average cost per year of owning a scooter vs. the average cost per year of owning a motorcycle.

Consider ways to decrease the cost of owning a scooter (engineer it to get higher gas mileage, lower the retail price, improve reliability so maintenance costs will go down).

Consider the ways to grow revenue (acquisition, steal share, grow the market, etc) -- the company's preference is to grow the market.

Consider expanding into new markets.

Solution

Cost of owning a motorcycle is currently slightly cheaper than owning a scooter. The actual cost will depend on assumptions about gas prices.

Look for various possible ideas for expanding the market from the candidate.

Candidate's Copy

Scooter and Motorcycle Data – 5 Years Ago Versus Today

Scooter Data		
	5 yrs ago	**Today**
Company Data		
Market Size (# motorcycles)	1,000,000	900,000
Market Share	70%	70%
Average Retail Price	$500	$500
Fixed Costs (per scooter)	$200	$200
Variable Costs (per scooter)	$200	$200
Customer Data		
Salvage Value	$0	$0
Average Useful Life (years)	4	4
Gas Mileage (mpg)	60	60
Average Annual Maintenance Cost	$100	$100

Motorcycle Data		
	5 yrs ago	**Today**
Company Data		
Market Size (# motorcycles)	100,000	500,000
Market Share	10%	10%
Average Retail Price	$600	$500
Fixed Costs (per scooter)	$200	$200
Variable Costs (per scooter)	$200	$200
Customer Data		
Salvage Value	$50	$50
Average Useful Life (years)	4	4
Gas Mileage (mpg)	60	60
Average Annual Maintenance Cost	$100	$100

Actual Case #3
Company: BCG

Case Description
A profitability analysis case.

Question
Heinz recently came up with the idea to double the volume of ketchup in a single-serve packet. How should the CEO approach this idea and what should he do?

Initial data to be given
None.

Additional data to be provided when asked
$500 million industry. Heinz has 60% of the market. Sells the product directly to large chains (McD's etc) and to food distributors (Sysco). The customers are entirely blind to the process - all they care about is that their cost stays low.
End users: Only 5% of customers only use 1 packet; 95% use 3-4; usage max out at 16 packets per sitting.
Competition: No one else is doing this, but would probably copy the idea if successful. So this idea won't be a source of competitive advantage.
Revenue: Expect it to be mostly flat - 60% of $500M market. Ketchup sold by the ton. Any large increase in amount purchased would be mostly offset by volume discount given.
Costs: 20% margins, so prod cost of $240M/year. Fixed costs are 50% of that ($120M) and are expected to be static. Variable costs: ketchup materials (50% of VC) savings of about 0.3% ($180K). Packet raw materials are other 50% of VC ($60M). Assume the packet is a 2x2 cube so volume = 8, surface area = 24.

Frameworks that might be used
Profitability

Solution
The issue here is production cost, specifically the variable cost of the ketchup packet raw materials. The candidate should be able to analyze the costs of producing the packets and clearly differentiate between those costs and the fact that the new packet will not give Heinz a competitive advantage. The candidate should be able to approximately calculate the surface area of the packet given the information above. If the size of contents is doubled, volume becomes 16. Still assume a cube and so each side of the cube is the cube root of 16 ~ 2.5. New surface area is now 2.5*2.5*6 = 37.5 (what the actual units are doesn't matter). Two of the old ketchup packets would have had surface area of 48, so have a savings of ~ 25% on materials with the bigger packet. 25% of $60M original cost is a savings of $15M. This is a good idea from Heinz's perspective, if it does not involve switching costs and they can get buy-in from their customers.

Actual Case #4
Company: BCG

Case Description
Growth strategy/operations strategy

Question
Mexican sewing machinery manufacturer is our client. Their industry is growing annually at 20%. Their sales, however, have been stagnant.

What is the cause of their stagnation, and what can they do?

Initial data to be given
Competition:
There have been new, foreign entrants into the marketplace. The existing players are small and irrelevant (they do not account for the growth). We have 35% of the market, and the foreigners have taken 10%.

Price:
The machines are priced differently: $100K vs. $150K.

Technology & Quality:
There is a difference between the quality of the sewing machines we manufacture and ones the foreign companies manufacture. All the machines have the same throughput, but the foreign machines produce less errors in their material: 1% vs. 1.5 to 2% in our client's. The cost of errors in manufacturing is $2 per unit of material.

Additional data to be provided when asked
Customers:
There is a difference in the customers (purchasers of machinery) between our client and their foreign competitors. Specifically, the customers of the foreigners are international themselves, and their quality standards are higher (they are the Wal-marts of the world, as opposed to local Mexican buyers). They buy the sewing machines locally and export them.

The growth in demand of machinery is entirely accounted for by foreign materials demand (buyers like Wal-mart) for higher-quality materials.

Currently, our client is not exporting any sewing machines.

Frameworks that might be used
Cost-benefit comparison followed by an analysis of the buyers' market (from Porter's)

Solution
A cost-benefit analysis shows that our machines are better: (throughput per day x days in a year x error differential x cost of error) > $150K - $100K.

It turns out that our client's customers are buying foreign machines because that is where the growing demand is (foreign markets).

Future strategy:
Interviewee should can talk about forming a relationship with Wal-Mart to export sewing machines. However, this area cannot be explored fully without knowing our production capacity.

Actual Case #5
Company: BCG

Case Description
An investment and profitability analysis case.

Question
Our client is a telecommunication company in Spain. It has the possibility of offering a new service called New-DSL.

Management is considering two options:

1) Offering it to both the retail division and competitors.
   This option would increase the market size by 50%.
2) Offering it only to its retail division (and not to competition)
   This option would increase the market size only by 20%, and the client's retail division would capture all the market.

The investment needed to go for this New-DSL technology would be 800€ millions.

You have been hired to recommend the client about whether pursue this New-DSL technology or keep the current DSL system. In case of going for the New-DSL, you should also suggest which option you think is better and why.

In addition to the financials, and regardless of the results, which option (between the two in the New-DSL system) is strategically better?

Initial data to be given
The client currently offers an Internet connection service based on DSL technology. This technology is mature and is in the fifth year after its launch. The company has two divisions, wholesale and retail.

In the retail segment, our client only has 40% of the market share, whereas competition has the remaining 60%.
In the distribution business, however, it has a monopoly, meaning that all retailers (including its own division) have to buy from them:

Distribution ----------- Retail division (40%) ----------- Final Client
-------------------- Competition (60%) ----------- Final Client

The current market size is 5 million customers.

Both wholesale and retail price are fixed by the law in 12€/month and 20€/month respectively. The wholesale cost is 10€/month, whereas the retail cost to provide the service is 7€/month. New-DSL technology would have the same cost/price structure as that of its existing DSL technology.

Additional data to be provided when asked
All the information has been provided at the beginning.

Frameworks that might be used
Profitability analysis, net present value (NPV) calculation.

Solution
The candidate should approach this case both from a financial and strategic perspective.

In order to proceed with the numbers, s/he has two possibilities: estimates either the profit from each option or the changes in profit the new-DSL would trigger.

Although the latter approach is taken as shown in the solutions, both approaches should yield the same results.

OPTION 1:

<u>Offer New-DSL to both retail division and competition</u>
50% of market increase (from 5M to 7.5M).
Since we have 40% of the market share, we already had 2M. Adding 40% of the new 2.5M, we would have 3M in total. The competition would capture 1.5M of the new market.

New profit for the company:

- Sale from wholesale division to competition.
  (12€ of revenue - 10€ of cost) x 1.5M = 3M € / month
  In 12 months ➔ 3x12 = 36M € / year
- Sale from wholesale to retail division
  We should not take into consideration this part because it is a transaction between divisions of the same company
- Sale from retail division to end user
  (20€ revenue - 17€ cost) x 1.0M users = 3.0M € / month
  In 12 months ➔ 3x12 = 36M € / year

TOTAL of **72M €** per year

We can assume a 5 year lifetime (the same the old DSL technology had):

Year	0	1	2	3	4	5
CF	-800	72	72	72	72	72

Regardless of the discount rate, this is clearly not a profitable business.

OPTION 2

<u>Offer New-DSL only to our retail division</u>

20% of market increase (from 5M to 6M)
Having 40% of the market share, we already had 2M. Since in this case, the company would capture all the market the new customer base is 6M (no leftover to competitors)
New profit for the company:

- Profit **lost** due to not selling to competition (we no longer sell 3M users to competition)
  (12€ of revenue - 10€ of cost) x 3M = 6M € / month
  In 12 months ➔ 6x12 = 72M €
- Sale from retail division to end user
  (20€ revenue - 17€ cost) x 4.0M users = 12.0M € / month
  In 12 months ➔ 14x12 = 144M €

TOTAL of **72M €** per year

Assuming the same 5 years of lifetime:

Year	0	1	2	3	4	5
CF	-800	72	72	72	72	72

Clearly, this is not profitable either.

Conclusion: After showing that neither option is profitable, the client should stick with its DSL technology as long as it manages the monopoly in the distribution business.

Strategically, however, and given the fact that the candidate has to choose between the two options for the New-DSL, option 2 could be interesting because we would expand our monopoly to the retail segment. However, it could imply legal implications. In addition, it could be dangerous because retails might join and create a distributor business and compete with us face-to-face in both segments.
Therefore, and given that both options are similar economically, option 1 would be more recommendable for the long run.

Actual Case #6
Company: McKinsey

Case Description
A quantitative case which requires: breakeven analysis, a market estimation, and recommendation of a long-term strategic position.

Question
Our client is a leading computer hardware manufacturer in a developing country. Its cell phone handset division is losing tens of millions of USD this year. It approaches us and wants to find out why and how to handle it.

Initial data to be given
1. The company wants to know whether it's achievable for them to at least break even in next year
2. Estimate the trend of the industry for coming years
3. Provide a recommendation for long-term

Additional data to be provided when asked
Company
1. The handset division is a new one and was established only last year. Last year they had reasonable profit.
2. Differentiation: The client has invested significantly in R&D but seems unable to differentiate itself.
3. Costs: The client has cost practice comparable to local competitors. Lack of scale is the main reason for their cost disadvantage. The client's cost structure for next year is expected as (illustrative):

Value Add	Cost	Per
Contracted parts	$50	Handset
Other parts	$20	Handset
Labor	$5	Handset
Utilities (direct & variable)	$3	Handset
Transportation	$4	Handset
SG&A	1,500,000	Annually
R&D	1,000,000	Annually
Depreciation	2,000,000	Annually

Market
The players in the handset market can be categorized into two groups:
- large multinationals such as Nokia, Sony-Ericsson and Motorola
- local suppliers.

The multinationals have been dominating power in the market for a long time; however, in recent years the local suppliers are beefing up in terms of both quality and design.

Leveraging their knowledge of local preference and strength in selling channeling, during the last two years local suppliers obtained roughly 20% of market share.

The market share landscape is like:

Company	Market Share
Nokia. Ericsson, Motorola and other Multinationals	80%
Local supplier A	5%
Local supplier B	5%
Local supplier C	3%
. . .	
Client (around 10th local supplier by MS)	<1%

Product

1. Price:
   - Handsets have become a commodity and price competition intensified.
   - Wholesale price (last year): $95
   - Wholesale price (this year): $90
   - Wholesale price (next year): 5% less (price war on-going)
2. Substitutes: There are some substitutes for the type of handset the client produces, but the threat is not significant.

Customers (client specific)

1. Two groups: Urban and Rural.

	Urban	Rural
Penetration rate (purchasing power differences)	50%	4%
Penetration growth rate (government has implemented supportive policies which is expected to grow market fast next year)	10%	50%
Average handset change	3 yrs	5 yrs
% of Population	30%	70%

2. Population of the country: 300M

Frameworks that might be used

3C, industry analysis, break-even calculation, market size calculation

Solution

1. Client needs to produce 1.3M units to breakeven

    Fixed Costs + Variable Costs * Units = Revenue/Unit * Units

Fixed Costs	= 1.5 + 1 + 2 = $4.5M
Variable Costs	= 50+20+5+3+4 = $82 per unit
Revenue	= $90 * 95% = $85.5 per unit
Contribution	= $3.5 per unit
Unit Breakeven	= 4.5 M / 3.5 per unit = 1.3M units

2. Market Size is ~ 25M units

    Urban Units

    Replacement Units = Population * % Urban * Penetration * 1 / Change

    300 * 30% * 50% * 1/3 = 15M

    New Units = Population * % Urban * Penetration * Growth

    300 * 30% * 50% * 10% = 4.5M

    Rural Units

    Replacement Units = Population * % Rural * Penetration * 1 / Change

    300 * 70% * 4% * 1/5 = 1.7M

    New Units = Population * % Urban * Penetration * Growth

    300 * 70% * 4% * 50% = 4.2M

    Total Units = Urban + Rural = 19.5M + 5.9M ~= 25.4M units

3. 1.3M units is about 5% of market

    = 1.3M / 25.4M =~ 5%

4. Since the client current has less than 1% market share, it will need to grow its share over five times next year in a commodity market to just break even.

5. Market growth was 50% or 8.7 M units. Client would need to capture 15% of this to breakeven.

    Growth = Change in Units / Last Yr Units = (25.4M – 16.7M) / 16.7 M = 50%

    Breakeven = 1.3M / 8.7M = 15%

6. Ideal Recommendations:
    a. Focus on high growth rural population with targeted marketing to capture at least 15% of total growth. Still not competitive cost position though.

AND/OR

    b. Consolidate with other players to achieve economy of scales (spreads fixed costs over more volume).

OR

    c. Exit (sell) if do not believe can capture growing market since cannot compete:
        i. Price: commodity market, need economies of scale (minor player for long-term sustainability)
        ii. Differentiation: R&D costs are high and still not differentiated.

Actual Case #7
Company: McKinsey

Case Description
A revenue analysis case

Question
Our client manages 260 parking lots in UK; these parking lots located in shopping malls, airports, railway stations and hospitals. The management team estimated revenue in 2006 would be 350M pounds.

Now the questions are:
1) How to justify whether the estimated revenue is reasonable?
2) How to increase revenue? What are the risk factors?

Initial data to be given
The case facts

Additional data to be provided when asked
In big cities, the utilization rate is high. For example, in London, the utilization rate is around 90%.

Frameworks that might be used
1. Revenue = Price * Volume
2. Demand/Supply
3. Price elasticity

Solution
1. Identify the source of revenue: Price and Volume.
2. What does the increased revenue come from? We need to consider new locations or higher utilization?
3. If the increased revenue is attributable to higher price, we should consider the elasticity of demand. When the demand is elastic, lower price will effectively boost demand. However, this promotion has limited help in cities because the high utilization rate.
4. Possible revenue enhancement ideas (test creativity): lower price, incentive program (ex. monthly pass), company contracts, marketing campaign or rent out some space for community activities.

Actual Case #8
Company: McKinsey

Case Description

Our client is an Indonesian bank, which was privatized three years ago. The bank focuses on retail banking and its clientele is the lower/middle class. The management team is planning regional expansion. Currently their target market is Singapore. They asked our advice regarding how to enter the market.

Question

1. What kind of customers they should pursue?
2. What kind of service they should provide?
3. Market Sizing
4. How to go after the target customers?

Initial data to be given

There are 3 local banks serving the general population and 6 foreign banks targeting affluent people. 80-90% of the Singapore population has a checking account.

Additional data to be provided when asked

People generally aren't interested in switching their deposit and checking account. They aren't interested in asset/wealth management. Because of high taxes, the demand for car purchase is low. (Asked for a number of possible services and narrowed the possibilities down to limited choices.) The lower income class of Singapore is about 30% of population.

Frameworks that might be used

1. Market Potential: market segmentation and service differentiation
2. Service Standardization

Solution

1. Customer: they should continue focusing on their expertise--service lower/middle income customers.
2. Use market research to identify "short-term loan" as the differentiating product. (A series of conversations with the interviewer)
3. Market sizing: total population* 30% (lower income class)*estimated credit line (average monthly income* X)* 18% (average APR for signature loans).
4. Similar to what has been done by Fannie Mae to the mortgage market. The bank can standardize the short-term loan products. The standard procedure can speed up the approval process and requires fewer employees and branches.
5. Potential risk: credit risk.